CLASS NUMBERS UP TO TEN THOUSAND AND ZERO-FREE HALF-PLANES

DEEP BHATTACHARJEE

Abstract. We prove, with every constant explicit, that a zero-free half-plane $\operatorname{Re} s > B$ for all Dirichlet L-functions gives two-sided bounds for $L(1,\chi)$ of Littlewood's shape with $\log\log q$ multiplied by $1/(1-B)$. For $B = 7/8$ these read $0.725/(8\log\log q - 4.15) \le |L(1,\chi)| \le 2.27\,(8\log\log q - 4.15)$ for conductors $q \ge 2^{40}$. Under this hypothesis every imaginary quadratic field of class number at most 10 798 has discriminant below 2^{40}, so its class number problem is solved by the existing unconditional tables, and we count the fields of each class number up to 1000. Without any hypothesis, Tatuzawa's theorem gives the same conclusion for class numbers up to 2900 with at most one possible exceptional field.

1. Introduction

Gauss conjectured in the *Disquisitiones Arithmeticae* that for each h only finitely many negative discriminants have class number h, and he gave the lists he believed complete for small h. The class number problem asks for these lists. Heilbronn proved finiteness, Siegel's theorem $L(1,\chi_d) \gg_\varepsilon |d|^{-\varepsilon}$ made it quantitative but ineffective [17], and the case $h = 1$ was settled by Heegner [6], Baker [1] and Stark [18]. Goldfeld's reduction [4] to an L-function with a zero of order three at the centre, supplied by Gross and Zagier [5], gave an effective but very small lower bound for $h(d)$, and Watkins [20] used it to determine every field with $h(d) \le 100$.[1]

The obstruction to doing better is the possible Siegel zero, a real zero of $L(s,\chi_d)$ very close to $s = 1$. A zero-free half-plane for all Dirichlet L-functions removes it. Proofs of such a half-plane have recently been announced: zero-freeness in $\operatorname{Re} s > 7/8$ [13], with a Lean formalisation of its Dirichlet part [15], the weaker region $\operatorname{Re} s > 11/12$ by a separate argument [14], and an improvement of $7/8$ to $0.874957\ldots$ by Liu [10]. These works have not yet been refereed, so in this paper the half-plane is a hypothesis, named below, and each theorem states it. A half-plane gives $L(1,\chi) \gg 1/\log\log q$ with an unstated constant. We make every constant explicit and certify it, and combine the result with the unconditional tabulation of Mosunov and Jacobson [12]; the outcome is the class number problem in a range a hundred times larger than that of Watkins, under the half-plane hypothesis.

For $\frac12 \le B < 1$ we write

$$\mathrm{H}(B): \qquad L(s,\chi) \ne 0 \ \text{ for every primitive Dirichlet character } \chi \text{ and every } s \text{ with } \operatorname{Re} s > B.$$

The announced theorem of [13] is H(7/8). Theorems 5.1 and 5.2 below are unconditional implications from H(B) to explicit bounds; the hypothesis enters only through them.

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Corresponding author: Deep Bhattacharjee, d.bhattacharjee@erl-forschung.de. For matters concerning the code and data: egspl@proton.me (GitHub: creelie).

[1]The Goldfeld–Gross–Zagier bound grows only like $\log|d|$, up to a factor depending on the prime divisors of d; Watkins combined it with extensive computations.

Theorem 1.1. *Assume* H(7/8). *Let χ be a primitive Dirichlet character of conductor $q \geq 2^{40}$. Then*

$$\frac{0.725}{8\log\log q - 4.15} \;\leq\; |L(1,\chi)| \;\leq\; 2.27\,(8\log\log q - 4.15).$$

Under the generalised Riemann hypothesis Littlewood [9] proved that $|L(1,\chi)|$ lies between constant multiples of $(\log\log q)^{-1}$ and $\log\log q$. Theorem 1.1 is the case $B = 7/8$ of Theorem 5.2, which gives such bounds under H(B) for every $B < 1$ with the factor $1/(1-B)$ in front of $\log\log q$. For the half-plane $\operatorname{Re} s > 11/12$ alone it gives $0.714/(12\log\log q - 8.55) \leq |L(1,\chi)|$.

For a fundamental discriminant $d < -4$ the class number formula reads $h(d) = \sqrt{|d|}\,L(1,\chi_d)/\pi$, where χ_d is the primitive real character of conductor $|d|$ [3, Chapter 6].

Theorem 1.2. *Assume* H(7/8). *Then every fundamental discriminant $d < -2^{40}$ has $h(d) \geq 10\,799$. Under the weaker hypothesis* H(11/12), *$h(d) \geq 7\,612$.*

Mosunov and Jacobson [12] computed, without any hypothesis, the class group of every imaginary quadratic field with $|d| < 2^{40}$; the smallest value of $L(1,\chi_d)$ in that range is $0.17070\ldots$, attained at $d = -107\,415\,709\,003$.

Corollary 1.3. *Assume* H(7/8), *and let $1 \leq h \leq 10\,798$. Every imaginary quadratic field of class number h has*

$$|d| \;\leq\; \Big(\frac{\pi h}{0.1706}\Big)^2 \;<\; 2^{40}.$$

In particular every such field has $|d| < 3.4 \cdot 10^{10}$ when $h \leq 10^4$, and the complete list of fields of class number h is the list contained in the tabulation of [12].

Proof. By Theorem 1.2, $|d| < 2^{40}$. By [12], $L(1,\chi_d) \geq 0.1706$ in that range, so $h = \sqrt{|d|}\,L(1,\chi_d)/\pi \geq 0.1706\sqrt{|d|}/\pi$ (the fields with $d = -3, -4$ have $h = 1$ and satisfy the bound). For $h \leq 10\,798$ the right-hand side of the display is $\leq 3.96 \cdot 10^{10} < 2^{40} = 1.0995 \cdot 10^{12}$. □

For $h \leq 1000$ the bound is $|d| \leq 3.3911 \cdot 10^8$, small enough to count reduced forms directly.

Theorem 1.4. *Assume* H(7/8). *For $1 \leq h \leq 1000$ the imaginary quadratic fields of class number h are exactly those with $|d| \leq 339\,110\,855$ and $h(d) = h$. There are $4\,115\,897$ of them; the largest $|d|$ is $227\,932\,027$, of class number 996. The number $F(h)$ of fields of class number h and the largest $|d|$ for each h are given in Appendix C for selected h and in full in the ancillary data.*

Without any hypothesis on zeros, the same tables decide the class number problem in a smaller range, up to a single field that cannot be excluded by present methods.

Theorem 1.5. *There is at most one fundamental discriminant $d^* < -2^{40}$ such that every fundamental discriminant $d < -2^{40}$ with $d \neq d^*$ satisfies $h(d) \geq 2901$. Consequently, for $1 \leq h \leq 2900$ every imaginary quadratic field of class number h, with the possible exception of the single field $\mathbb{Q}(\sqrt{d^*})$, has $|d| \leq (\pi h/0.1706)^2$ and appears in the tabulation of* [12]*; in particular the counts of Theorem 1.4 are unconditionally complete apart from at most one field.*

Theorem 1.5 is proved in Section 6. It uses no zero-free region; the price is the possible exception, which is the Siegel zero in another form.

For $h \leq 100$ the counts and the largest discriminants agree with those of Watkins [20]; his tables stop at $h = 100$.

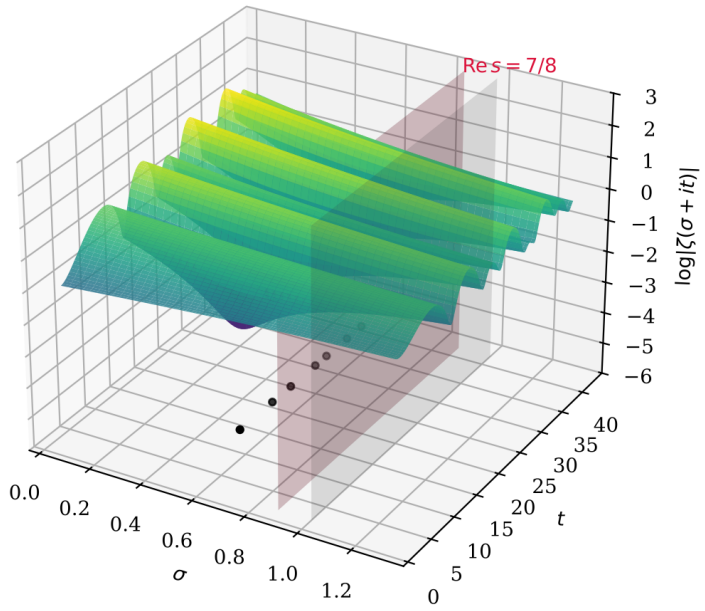


FIGURE 1. The surface $\log|\zeta(\sigma + it)|$ over the critical strip, with the plane $\sigma = 7/8$ and the first zeros on the line $\sigma = 1/2$. The theorems of this paper use only that no zero of any Dirichlet L-function lies to the right of the plane.

Method. The proof of Theorem 1.1 is a smoothed explicit formula. Integrating the formula for $\sum \Lambda_\chi(n) n^{-1-u} \log(x/n)$ over $u \in [0, \infty)$ produces $\log L(1, \chi)$ itself on one side and, on the other, a sum over zeros in which each zero ρ contributes at most $x^{\beta-1}/(|\rho - 1|^2 \log x)$. Taking the difference of two such formulas at x and y leaves a prime sum with a weight that is 1 up to y and decays linearly in $\log n$ up to x (Figure 2). Under $\mathrm{H}(B)$ every term $x^{\beta-1}$ is at most x^{B-1}, the sum $\sum_\rho |\rho - 1|^{-2}$ is $O(\log q)$ by the Hadamard product, and the choice $y = (\log q)^{1/(1-B)}$ makes the zero contribution bounded. The prime sum is then bounded with the explicit estimates of Rosser and Schoenfeld [16]. Every constant is evaluated in interval arithmetic [7].

Organisation. Section 2 proves the explicit formula and its integrated form, Section 3 bounds the zeros, Section 4 the primes, and Section 5 proves Theorems 1.1 and 1.2. Section 6 proves Theorem 1.5 and Section 7 proves Theorem 1.4. Appendix A contains the contour argument, Appendix B the certified constants, Appendix C the data.

2. The integrated explicit formula

Throughout, χ is a primitive character modulo $q \geq 3$, $a \in \{0, 1\}$ is defined by $\chi(-1) = (-1)^a$, $\Lambda_\chi(n) = \Lambda(n)\chi(n)$, and $\rho = \beta + i\gamma$ runs over the nontrivial zeros of $L(s, \chi)$ with multiplicity. They satisfy $0 < \beta < 1$, and $\sum_\rho |\rho|^{-2} < \infty$ [3, Chapter 12]. The trivial zeros are $s = -2k - a$, $k \geq 0$.

Lemma 2.1. *For $x > 1$ and $u \geq 0$,*

$$\sum_{n\leq x} \frac{\Lambda_\chi(n)}{n^{1+u}} \log\frac{x}{n} = -\frac{L'}{L}(1+u,\chi)\log x - \Big(\frac{L'}{L}\Big)'(1+u,\chi) - \sum_{\rho} \frac{x^{\rho-1-u}}{(\rho-1-u)^2} - \sum_{k\geq 0} \frac{x^{-2k-a-1-u}}{(2k+a+1+u)^2}. \tag{1}$$

Both sums converge absolutely.

Proof. Put $s = 1+u$. For $y > 0$ and $c > 0$ one has $\frac{1}{2\pi i}\int_{(c)} y^w w^{-2}\,dw = \max(\log y, 0)$, with absolute convergence. Since $-\frac{L'}{L}(s+w,\chi) = \sum \Lambda_\chi(n) n^{-s-w}$ converges absolutely on $\operatorname{Re} w = 1$, termwise integration gives

$$\sum_{n\leq x} \frac{\Lambda_\chi(n)}{n^s} \log\frac{x}{n} = \frac{1}{2\pi i}\int_{(1)} \Big(-\frac{L'}{L}(s+w,\chi)\Big)\frac{x^w}{w^2}\,dw.$$

The integrand is meromorphic. At $w = 0$ it has a double pole with residue $-\frac{L'}{L}(s,\chi)\log x - (\frac{L'}{L})'(s,\chi)$; at $w = \rho - s$ a pole with residue $-x^{\rho-s}/(\rho-s)^2$ times the multiplicity of ρ; at $w = -2k-a-s$ a pole with residue $-x^{-2k-a-s}/(2k+a+s)^2$. Moving the line of integration to $-\infty$ through a sequence of rectangles on which L'/L is controlled (Appendix A) proves (1). Absolute convergence follows from $|\rho - s| \geq |\rho - 1| \gg |\rho|$ and $\sum |\rho|^{-2} < \infty$.[2] □

Integration of (1) over u removes the derivative and leaves $\log L(1,\chi)$. Here $\log L(s,\chi) = \sum_{n\geq 2} \Lambda_\chi(n) n^{-s}/\log n$ for $\operatorname{Re} s > 1$, continued along the real segment to $s = 1$; this is possible because $L(1+u,\chi) \neq 0$ for $u \geq 0$.

Proposition 2.2. *For $x > 1$ put*

$$V(x) = \sum_{2\leq n\leq x} \frac{\Lambda_\chi(n)}{n\log n}\log\frac{x}{n}.$$

Then

$$V(x) = \log L(1,\chi)\,\log x + \frac{L'}{L}(1,\chi) - \sum_{\rho} I_\rho(x) - T(x),$$

where

$$|I_\rho(x)| \leq \frac{x^{\beta-1}}{|\rho-1|^2\log x}, \qquad |T(x)| \leq \frac{\pi^2}{8\,x\log x}.$$

Proof. Integrate (1) over $u \in [0,\infty)$. On the left, $\int_0^\infty n^{-1-u}\,du = 1/(n\log n)$ for $n \geq 2$, which gives $V(x)$. On the right, $\int_0^\infty -\frac{L'}{L}(1+u,\chi)\,du = \log L(1,\chi)$ and $\int_0^\infty -(\frac{L'}{L})'(1+u,\chi)\,du = \frac{L'}{L}(1,\chi)$, since $\log L(1+u,\chi)$ and $\frac{L'}{L}(1+u,\chi)$ tend to 0 as $u \to \infty$. The zero term is

$$I_\rho(x) = \int_0^\infty \frac{x^{\rho-1-u}}{(\rho-1-u)^2}\,du.$$

Since $1-\beta+u \geq 1-\beta > 0$, we have $|\rho-1-u|^2 = (1-\beta+u)^2 + \gamma^2 \geq |\rho-1|^2$, hence

$$|I_\rho(x)| \leq \frac{x^{\beta-1}}{|\rho-1|^2}\int_0^\infty x^{-u}\,du = \frac{x^{\beta-1}}{|\rho-1|^2\log x}.$$

[2]The square of w in the kernel x^w/w^2 is what makes the sum over zeros converge absolutely, with no truncation in height.

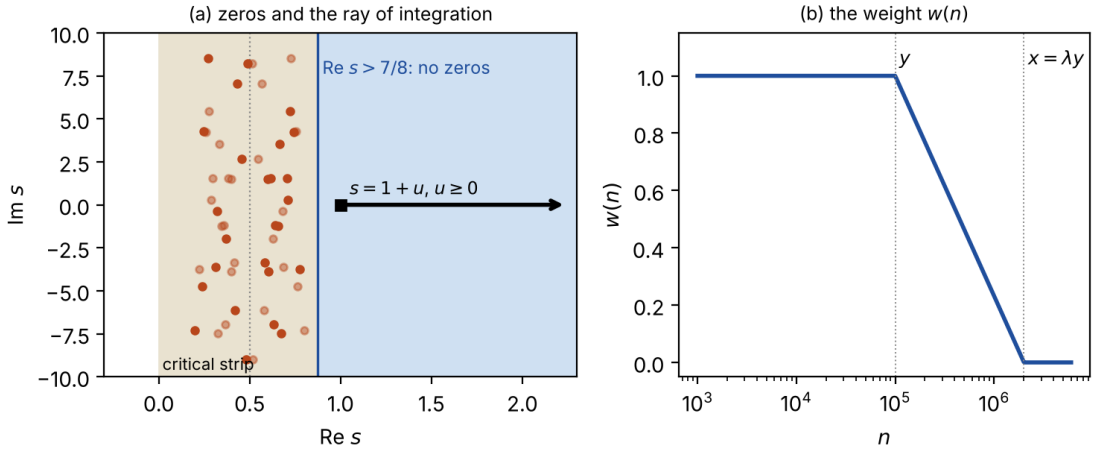


FIGURE 2. (a) Under H(7/8) every zero lies in $\beta \leq 7/8$; the formula is integrated along the ray $s = 1 + u$. (b) The weight of Corollary 2.3: equal to 1 up to y and linear in $\log n$ between y and $x = \lambda y$.

In the same way the k-th trivial term is at most $x^{-2k-a-1}/((2k+a+1)^2 \log x)$, and $\sum_{k\geq 0}(2k+a+1)^{-2} \leq \sum_{k\geq 0}(2k+1)^{-2} = \pi^2/8$. Interchanging the integral with the sums over zeros is justified by the bounds just proved and $\sum_\rho |\rho - 1|^{-2} < \infty$. □

Subtracting the formula at y from the formula at x eliminates the unknown constant $\frac{L'}{L}(1,\chi)$.

Corollary 2.3. *Let $1 < y < x$, $\lambda = x/y$ and $\ell = \log \lambda$. Define $w(n) = 1$ for $n \leq y$, $w(n) = \log(x/n)/\ell$ for $y < n \leq x$, and $w(n) = 0$ for $n > x$. Then*

$$\log L(1,\chi) = \sum_{n\geq 2} \frac{\Lambda_\chi(n)\, w(n)}{n \log n} + \theta\, \mathcal{E}, \qquad |\theta| \leq 1, \tag{2}$$

where

$$\mathcal{E} = \frac{1}{\ell}\Big[\Big(\frac{x^{\beta^*-1}}{\log x} + \frac{y^{\beta^*-1}}{\log y}\Big) \sum_\rho \frac{1}{|\rho - 1|^2} + \frac{\pi^2}{8}\Big(\frac{1}{x \log x} + \frac{1}{y \log y}\Big)\Big]$$

and $\beta^* = \sup_\rho \beta$.

Proof. For $n \leq y$ the difference $\log(x/n) - \log(y/n)$ equals ℓ; for $y < n \leq x$ only $\log(x/n)$ is present. Hence $V(x) - V(y) = \ell \sum \Lambda_\chi(n) w(n)/(n \log n)$. Proposition 2.2 gives $V(x) - V(y) = \ell \log L(1,\chi) - \sum_\rho (I_\rho(x) - I_\rho(y)) - (T(x) - T(y))$, and the bounds there with $x^{\beta-1} \leq x^{\beta^*-1}$ give the claim. □

3. The sum over zeros

Lemma 3.1. *Assume that every zero of $L(s,\chi)$ has $\beta \leq B$, where $\frac{1}{2} \leq B < 1$. Then for every $u > 0$,*

$$\sum_\rho \frac{1}{|\rho - 1|^2} \;\leq\; S(q,u,B) := \frac{1 + u - B}{(1-B)^2}\Big(\frac{1}{2}\log\frac{q}{\pi} + \frac{1}{2}\psi\Big(1 + \frac{u}{2}\Big) - \frac{\zeta'}{\zeta}(1+u)\Big),$$

where $\psi = \Gamma'/\Gamma$.

Proof. Let $\sigma = 1+u$. The Hadamard product [3, Chapter 12] gives

$$\frac{L'}{L}(s,\chi) = -\frac{1}{2}\log\frac{q}{\pi} - \frac{1}{2}\psi\Big(\frac{s+a}{2}\Big) + b(\chi) + \sum_\rho\Big(\frac{1}{s-\rho}+\frac{1}{\rho}\Big), \qquad \operatorname{Re} b(\chi) = -\sum_\rho \operatorname{Re}\frac{1}{\rho},$$

so that, at $s=\sigma$,

$$\sum_\rho \frac{\sigma-\beta}{|\sigma-\rho|^2} = \frac{1}{2}\log\frac{q}{\pi} + \frac{1}{2}\psi\Big(\frac{\sigma+a}{2}\Big) + \operatorname{Re}\frac{L'}{L}(\sigma,\chi). \tag{3}$$

Since ψ is increasing, $\psi((\sigma+a)/2) \le \psi(1+u/2)$, and $|\frac{L'}{L}(\sigma,\chi)| \le \sum \Lambda(n)n^{-\sigma} = -\frac{\zeta'}{\zeta}(\sigma)$. On the other side, $\sigma-\beta \ge 1-\beta > 0$ gives $|\sigma-\rho|^2 \le ((\sigma-\beta)/(1-\beta))^2|1-\rho|^2$, that is,

$$\frac{1}{|1-\rho|^2} \le \frac{\sigma-\beta}{(1-\beta)^2}\cdot\frac{\sigma-\beta}{|\sigma-\rho|^2}.$$

The function $\beta \mapsto (\sigma-\beta)/(1-\beta)^2$ is increasing on $(-\infty,1)$, because its derivative is $(1-\beta)^{-3}(2\sigma-1-\beta) > 0$, so it is at most $(\sigma-B)/(1-B)^2$. Summing over ρ and using (3) proves the lemma. □

Remark 3.2. The factor $(1+u-B)/(1-B)^2$ is where the width of the zero-free region enters. Under the Riemann hypothesis for $L(s,\chi)$ it would be $4(1/2+u)$, and the bound would be the classical $\sum_\rho |\rho-1|^{-2} = \sum_\rho 1/(1/4+\gamma^2) \sim \log q$.

4. The sum over primes

Let $B_1 = 0.2614972128\ldots$ be Mertens' constant, so that $\sum_{p\le t} 1/p = \log\log t + B_1 + o(1)$, let γ be Euler's constant, and put

$$G(y) = \sum_{p\le y}\Big(\frac{1}{p} - \log\Big(1+\frac{1}{p}\Big)\Big), \qquad F(v) = v\log v - v + B_1 v - \frac{1}{2v}.$$

The terms of G are positive, so G increases; $G(10^6) = 0.1819818165\ldots$ and $G(\infty) = 0.18198\ldots$.

Lemma 4.1. *Let $286 \le y < x$, $\ell = \log(x/y)$, and let M be the sum in (2). Put $W = (F(\log x) - F(\log y))/\ell$. Then*

$$-W + G(y) - \frac{3}{\sqrt{y}} \;\le\; \operatorname{Re} M \;\le\; W + \gamma - B_1 + \frac{3}{\sqrt{y}}.$$

Proof. Only prime powers contribute, and $\Lambda(p^k)/(p^k\log p^k) = 1/(kp^k)$. Since $w(p^k) = 1$ when $p^k \le y$, we can write

$$M = \sum_{p\le y} -\log\Big(1-\frac{\chi(p)}{p}\Big) + \sum_{y<p\le x}\frac{\chi(p)w(p)}{p} + R,$$

where

$$R = \sum_{\substack{p\le y,\ k\ge 2\\ p^k>y}} \frac{\chi(p)^k(w(p^k)-1)}{kp^k} + \sum_{\substack{p>y\\ k\ge 2}}\frac{\chi(p)^k w(p^k)}{kp^k}.$$

Since $0 \le w \le 1$, we have $|R| \le \frac{1}{2}\sum_p \sum_{k\ge 2,\, p^k>y} p^{-k}$. For fixed p let k_0 be the least $k \ge 2$ with $p^k > y$; the inner sum is at most $p^{-k_0}p/(p-1) \le 2p^{-k_0}$. If $p \le \sqrt{y}$ then $p^{-k_0} < 1/y$, and there are at most $\sqrt{y}$ such primes; if $p > \sqrt{y}$ then $k_0 = 2$. Hence

$$|R| \le \frac{\sqrt{y}}{y} + \sum_{m>\sqrt{y}}\frac{1}{m^2} \le \frac{1}{\sqrt{y}} + \frac{1}{\sqrt{y}-1} \le \frac{3}{\sqrt{y}}.$$

For the lower bound, $\operatorname{Re}(-\log(1-\chi(p)/p)) = -\log|1-\chi(p)/p| \ge -\log(1+1/p)$ and $\operatorname{Re}\chi(p)w(p)/p \ge -w(p)/p$. Thus

$$\operatorname{Re} M \ge G(y) - \Big(\sum_{p\le y}\frac{1}{p} + \sum_{y<p\le x}\frac{w(p)}{p}\Big) - \frac{3}{\sqrt{y}}.$$

With $P(t)=\sum_{p\le t}1/p$, partial summation gives the exact identity

$$\sum_{p\le y}\frac{1}{p} + \sum_{y<p\le x}\frac{\log(x/p)}{\ell\, p} = \frac{1}{\ell}\int_y^x P(t)\,\frac{dt}{t}.$$

By Rosser and Schoenfeld [16, Theorem 5], $P(t) < \log\log t + B_1 + 1/(2\log^2 t)$ for $t \ge 286$. The substitution $v=\log t$ turns the integral into $\ell^{-1}\int_{\log y}^{\log x}(\log v + B_1 + \frac{1}{2v^2})\,dv = W$. This proves the lower bound. For the upper bound we use $-\log|1-\chi(p)/p| \le -\log(1-1/p)$ and Mertens' identity[3]

$$\sum_p\Big(-\log\Big(1-\frac{1}{p}\Big)-\frac{1}{p}\Big) = \gamma - B_1,$$

whose terms are positive, so that $\sum_{p\le y} -\log(1-1/p) \le P(y)+\gamma-B_1$. The rest of the argument is the same. □

5. Proofs of Theorems 1.1 and 1.2

Combining Corollary 2.3 with Lemmas 3.1 and 4.1 gives a bound valid for each choice of parameters.

Theorem 5.1. *Assume* H(B) *with* $\frac12 \le B < 1$. *Let* χ *be primitive of conductor* $q \ge 3$, *let* $u>0$, $286 \le y < x$, $\ell = \log(x/y)$, *and let* W *be as in Lemma 4.1. Put*

$$E = \frac{1}{\ell}\Big[\Big(\frac{x^{B-1}}{\log x} + \frac{y^{B-1}}{\log y}\Big)S(q,u,B) + \frac{\pi^2}{8}\Big(\frac{1}{x\log x}+\frac{1}{y\log y}\Big)\Big].$$

Then

$$-W + G(y) - \frac{3}{\sqrt{y}} - E \;\le\; \log|L(1,\chi)| \;\le\; W + \gamma - B_1 + \frac{3}{\sqrt{y}} + E.$$

Proof. Take real parts in (2): $\log|L(1,\chi)| = \operatorname{Re} M + \operatorname{Re}\theta\mathcal{E}$. Under H($B$), $\beta^* \le B$, and Lemma 3.1 gives $\mathcal{E} \le E$. Lemma 4.1 bounds $\operatorname{Re} M$. □

For a single conductor the parameters can be optimised. At $q = 2^{40}$ and $B = 7/8$ the best choice found is $y = 10^{6.65}$, $\ell = 14$, $u = 1/10$, which gives $L(1,\chi) \ge 0.032934$ (Figure 3). To cover all q at once we let y grow with q.

Theorem 5.2. *Assume* H(B) *with* $\frac12 \le B < 1$ *and put* $k = 1/(1-B)$. *Let* $q_0 \ge 3$, $y_0 \ge 10^6$, $\ell > 0$ *and* $u > 0$, *and suppose that*

$$c_0 = \frac12\psi\Big(1+\frac{u}{2}\Big) - \frac12\log\pi - \frac{\zeta'}{\zeta}(1+u) > 0.$$

[3]It follows from Mertens' theorems $\prod_{p\le t}(1-1/p)^{-1} \sim e^\gamma \log t$ and $\sum_{p\le t}1/p = \log\log t + B_1 + o(1)$ by taking logarithms; see [11, Theorem 2.7].

Let E_0 be the quantity E of Theorem 5.1 at $q = q_0$, $y = y_0$, $x = e^{\ell} y_0$, and put

$$\kappa_0 = \log y_0 + \frac{\ell}{2} - k \log\log q_0,$$
$$C_- = \exp\Big(G(10^6) - B_1 - \frac{1}{2\log^2 y_0} - \frac{3}{\sqrt{y_0}} - E_0\Big),$$
$$C_+ = \exp\Big(\gamma + \frac{1}{2\log^2 y_0} + \frac{3}{\sqrt{y_0}} + E_0\Big).$$

Then every primitive χ of conductor $q \geq q_0$ satisfies

$$\frac{C_-}{k\log\log q + \kappa_0} \leq |L(1,\chi)| \leq C_+ \,(k\log\log q + \kappa_0).$$

Proof. Apply Theorem 5.1 with

$$y = y_0\Big(\frac{\log q}{\log q_0}\Big)^k, \qquad x = e^{\ell} y.$$

Then $y \geq y_0$ and $\log y + \ell/2 = k\log\log q + \kappa_0 > 0$.

The zero term. Write $L = \log q$ and $L_0 = \log q_0$. Since $(B-1)k = -1$, we have $y^{B-1} = y_0^{B-1} L_0/L$ and $x^{B-1} = e^{\ell(B-1)} y^{B-1}$. With $A = (1+u-B)/(1-B)^2$, $S(q,u,B) = A(\frac{1}{2}L + c_0)$. Therefore

$$y^{B-1} S(q,u,B) = y_0^{B-1} L_0 A\Big(\frac{1}{2} + \frac{c_0}{L}\Big),$$

which decreases in q because $c_0 > 0$; the same holds with x in place of y. As $\log x$, $\log y$, x and y increase with q, every term of E decreases, so $E \leq E_0$.

The prime term. The function $\log v$ is concave, so $\ell^{-1}\int_{\log y}^{\log x} \log v\, dv \leq \log(\log y + \ell/2)$; also $\ell^{-1}\int_{\log y}^{\log x} dv/(2v^2) \leq 1/(2\log^2 y)$. Hence

$$W \leq \log(k\log\log q + \kappa_0) + B_1 + \frac{1}{2\log^2 y_0}.$$

Finally $G(y) \geq G(10^6)$ and $3/\sqrt{y} \leq 3/\sqrt{y_0}$. Inserting these bounds into Theorem 5.1 and exponentiating gives the result. □

Proof of Theorem 1.1. By assumption H(7/8) holds. Take $q_0 = 2^{40}$, $y_0 = 5\cdot 10^6$, $\ell = 14$, $u = 1/10$. The certified values (Appendix B) are $c_0 = 8.6197\ldots > 0$, $E_0 = 0.237876\ldots$, $\kappa_0 = -4.153984\ldots$, $C_- = 0.725543\ldots$ and $C_+ = 2.267175\ldots$. Since $\kappa_0 < -4.15$, $C_- > 0.725$ and $C_+ < 2.27$, Theorem 5.2 gives Theorem 1.1. □

Proof of Theorem 1.2. Let $d < -2^{40}$ be fundamental and $q = |d|$. Theorem 5.2, with the constants of the previous proof, gives

$$h(d) = \frac{\sqrt{q}\, L(1,\chi_d)}{\pi} \geq \frac{C_-\sqrt{q}}{\pi\,(8\log\log q + \kappa_0)}.$$

The right-hand side increases with q for $q \geq 2^{40}$, since the derivative of log of it is $\frac{1}{2q} - \frac{8}{q\log q\,(8\log\log q + \kappa_0)} > 0$ there. At $q = 2^{40}$ it equals $10\,798.96\ldots$, so $h(d) \geq 10\,799$. For $B = 11/12$ take $y_0 = 1.8\cdot 10^9$, $\ell = 20$, $u = 0.08$; then $k = 12$, $\kappa_0 = -8.5573\ldots$, $C_- = 0.71408\ldots$, and the same computation gives $h(d) \geq 7\,612.06\ldots$. □

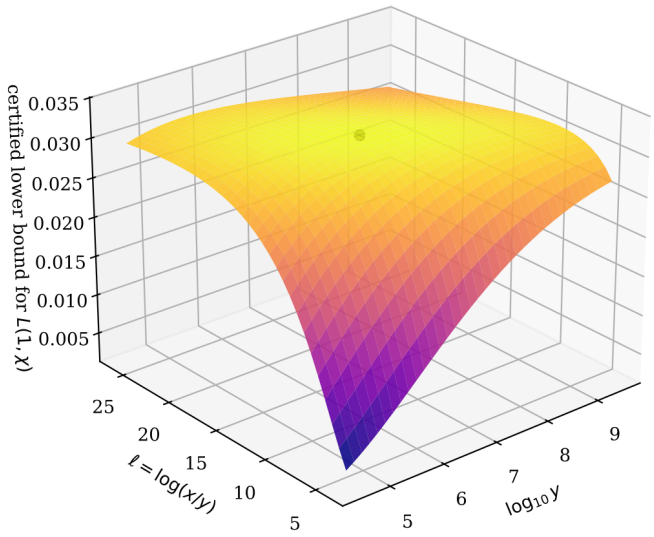


FIGURE 3. The certified lower bound for $L(1,\chi)$ at $q = 2^{40}$ given by Theorem 5.1 under H(7/8), as a function of $\log_{10} y$ and ℓ, at $u = 1/10$. The maximum, 0.032934, lies near $(6.65, 14)$.

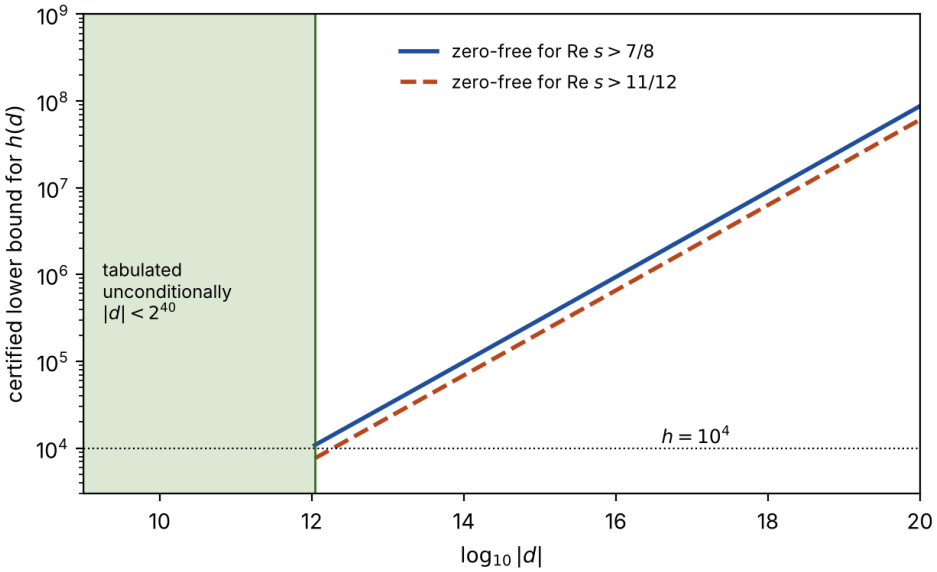


FIGURE 4. The lower bound for $h(d)$ of Theorem 1.2 for $|d| \geq 2^{40}$, from the half-planes $\operatorname{Re} s > 7/8$ and $\operatorname{Re} s > 11/12$. To the left of 2^{40} every class group is known [12].

Remark 5.3. Liu's half-plane $\operatorname{Re} s > 0.874957\ldots$ [10] changes the optimised bound at $q = 2^{40}$ by less than 10^{-5}. The constant that matters is $k = 1/(1-B)$: as $B \to \frac{1}{2}$ the method returns

bounds of Littlewood's order with $k = 2$. Explicit forms of Littlewood's bounds under the generalised Riemann hypothesis were given by Lamzouri, Li and Soundararajan [8].

Remark 5.4. For a real quadratic field of discriminant $d > 0$ the class number formula $h(d)\log\varepsilon_d = \sqrt{d}\,L(1,\chi_d)/2$ and Theorem 1.1 give $h(d)\log\varepsilon_d \geq 0.36\sqrt{d}/(8\log\log d - 4.15)$ for $d \geq 2^{40}$.

6. An unconditional range with one possible exception

Tatuzawa [19, Lemma 9 and Theorem 2] proved the following effective form of Siegel's theorem: if $0 < \varepsilon < 1/11.2$, then with at most one exception every fundamental discriminant D with $|D| \geq \max(e^{1/\varepsilon}, e^{11.2})$ satisfies

$$L(1,\chi_D) \;\geq\; 0.655\,\varepsilon\,|D|^{-\varepsilon}. \tag{4}$$

Proof of Theorem 1.5. Take $\varepsilon = 1/\log(2^{40}) = 0.036067\ldots$, which is below $1/11.2$, so that $e^{1/\varepsilon} = 2^{40} > e^{11.2}$. Let d^* be the exceptional discriminant of (4) for this ε, if there is one. For every other fundamental $d < -2^{40}$ the class number formula and (4) give

$$h(d) = \frac{\sqrt{|d|}\,L(1,\chi_d)}{\pi} \;\geq\; \frac{0.655\,\varepsilon}{\pi}\,|d|^{1/2-\varepsilon} \;\geq\; \frac{0.655\,\varepsilon}{\pi}\,\bigl(2^{40}\bigr)^{1/2-\varepsilon} = \frac{0.655\cdot 2^{20}}{e\,\pi\log 2^{40}} = 2900.757\ldots,$$

because $1/2 - \varepsilon > 0$ and $(2^{40})^{-\varepsilon} = e^{-1}$. Hence $h(d) \geq 2901$. If $h \leq 2900$ and $d \neq d^*$ has class number h, then $|d| < 2^{40}$, and the argument of Corollary 1.3, which uses only [12] once $|d| < 2^{40}$ is known, gives $|d| \leq (\pi h/0.1706)^2$. For $h \leq 1000$ this is the range counted in Theorem 1.4. □

The possible exception reflects a possible real zero of $L(s,\chi_{d^*})$ very close to 1, which the hypothesis H(B) excludes. An unconditional effective bound $L(1,\chi_d) > 10^4\pi/\sqrt{|d|}$ for all fundamental $d < -2^{40}$ would remove it in the range $h \leq 10^4$; no such bound is known, and the best unconditional effective lower bound for $h(d)$, from the method of Goldfeld, Gross and Zagier, grows only like $\log|d|$.

7. Class numbers up to one thousand

Let $D > 4$ and $d = -D$. A binary quadratic form $ax^2 + bxy + cy^2$ with $b^2 - 4ac = d$ and $a > 0$ is *reduced* if $|b| \leq a \leq c$, and $b \geq 0$ whenever $|b| = a$ or $a = c$. Every primitive positive definite form is properly equivalent to exactly one reduced form [2, Theorem 2.8], and the classes of primitive forms of discriminant d form a group isomorphic to the class group of the order of discriminant d [2, Theorem 7.7]. If d is fundamental then every form of discriminant d is primitive, since a common factor $g > 1$ of a, b, c would make d/g^2 a discriminant. Thus for fundamental d,

$$h(d) = \#\{(a,b,c) \in \mathbb{Z}^3 :\; b^2 - 4ac = d,\;\; (a,b,c) \text{ reduced}\}.$$

A reduced form has $3a^2 \leq 4ac - b^2 = D$, so $a \leq \sqrt{D/3}$.

Proof of Theorem 1.4. Let $h(d) \leq 1000$. By Corollary 1.3, $|d| \leq (1000\pi/0.1706)^2 < N := 339\,110\,855$. We computed $h(d)$ for every fundamental d with $|d| \leq N$ by counting reduced forms: for each $a \leq \sqrt{N/3}$ and $-a < b \leq a$, every $c \geq a$ (with $c > a$ when $b < 0$) with $4ac - b^2 \leq N$ adds one to the count of $D = 4ac - b^2$. The discriminant $-D$ is fundamental if and only if $D \equiv 3 \pmod 4$ and D is squarefree, or $D = 4m$ with $m \equiv 1, 2 \pmod 4$ and m squarefree. The program and its output are in the ancillary files. The counts for $h \leq 100$ agree, for every h, with [20]. □

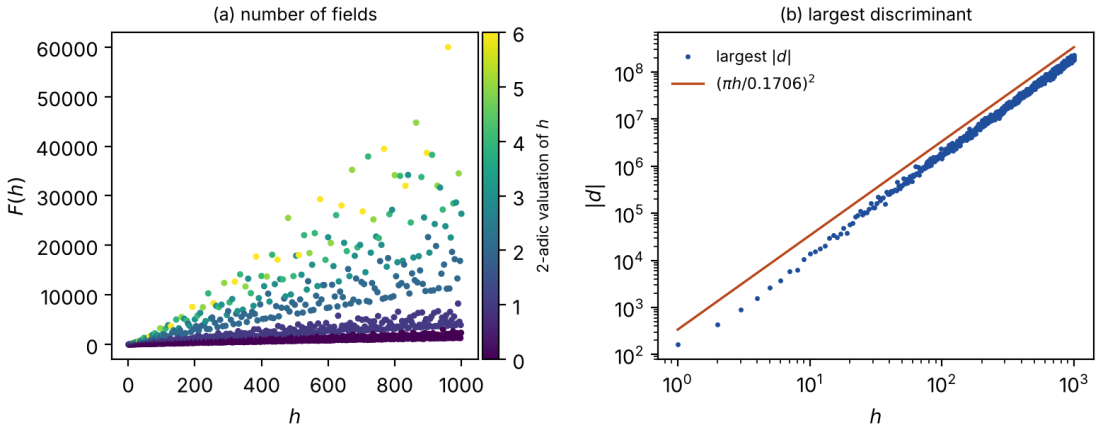


FIGURE 5. (a) The number $F(h)$ of imaginary quadratic fields of class number h, for $h \le 1000$; the upper points are the h divisible by a high power of 2. (b) The largest $|d|$ of class number h, with the bound $(\pi h/0.1706)^2$ of Corollary 1.3.

The computation took 15 minutes on four cores. Within $|d| \le N$ it finds 103 077 319 fundamental discriminants, the smallest value of $L(1,\chi_d) = \pi h(d)/\sqrt{|d|}$ being 0.198814 at $d = -991\,027$. Figure 5 shows $F(h)$ and the largest $|d|$ against h.

Remark 7.1. For $1000 < h \le 10\,798$ the same counting is possible in principle, but $|d|$ then reaches $3.9 \cdot 10^{10}$, and the number of operations grows like $N^{3/2}$. Corollary 1.3 shows that the answer is contained in the tabulation of Mosunov and Jacobson [12], which covers $|d| < 2^{40}$ with all class groups computed unconditionally.[4]

Remark 7.2. The Riemann hypothesis itself is not touched by this work. It would replace $k = 8$ by $k = 2$ in Theorem 1.1. Any unconditional zero-free half-plane $\operatorname{Re} s > B$ with $B < 1$ makes all results of this paper unconditional, with the constants of Theorem 5.2 for that B.

APPENDIX A. THE CONTOUR ARGUMENT

We justify the shift of contour in Lemma 2.1. Put $s = 1 + u$ and write $s' = s + w$. We use three facts [3, Chapters 9, 12, 16, 17 and 19], together with $|\frac{L'}{L}(\sigma + it, \chi)| \le -\frac{\zeta'}{\zeta}(2)$ for $\sigma \ge 2$.

(1) For each $j \ge 2$ there is $T_j \in [j, j+1]$ such that $|\frac{L'}{L}(\sigma + iT_j, \chi)| \ll \log^2(qT_j)$ uniformly for $-1 \le \sigma \le 2$; the same holds for $-T_j$.
(2) For $\sigma \le -1$ and s' at distance at least $\frac{1}{2}$ from every trivial zero, $|\frac{L'}{L}(s', \chi)| \ll \log(q|s'|)$, by the functional equation and Stirling's formula.
(3) The number of zeros with $|\gamma - t| \le 1$ is $O(\log(q(|t| + 2)))$.

Let $m \ge 1$ and let $\mathcal{R}$ be the rectangle with vertices $1 \pm iT_j$ and $-U_m \pm iT_j$, where $U_m = s + 2m + a + 1$, so that the left side lies on $\operatorname{Re} s' = -2m - a - 1$, midway between two trivial zeros. By the residue theorem the integral over the right side of $\mathcal{R}$ equals the sum of the residues inside $\mathcal{R}$ plus the integrals over the other three sides. On the horizontal sides, $|x^w/w^2| \le x^{\operatorname{Re} w}/T_j^2$

[4]For $d \not\equiv 1 \pmod 8$ they use class number formulas for theta series; for $d \equiv 1 \pmod 8$ they use the batch verification of Jacobson, Ramachandran and Williams, based on the Eichler–Selberg trace formula, which removes the dependence on the extended Riemann hypothesis.

and, by (1), (2) and the bound for $\sigma \geq 2$, $|\frac{L'}{L}(s', \chi)| \ll \log^2(q(T_j + U_m + 2))$; the contribution is

$$\ll \frac{\log^2(q(T_j + U_m + 2))}{T_j^2} \int_{-\infty}^{1} x^v \, dv \ll \frac{x \log^2(q(T_j + U_m + 2))}{T_j^2 \log x} \longrightarrow 0 \qquad (j \to \infty, \ m \text{ fixed}).$$

On the left side, (2) gives the bound

$$\ll x^{-U_m} \int_{-\infty}^{\infty} \frac{\log(q(U_m + |t| + 2))}{U_m^2 + t^2} \, dt \ll x^{-U_m} \frac{\log(qU_m)}{U_m} \longrightarrow 0 \qquad (m \to \infty).$$

Letting first $j \to \infty$ and then $m \to \infty$, and using (3) and $|\rho - s| \geq |\rho - 1|$ to see that the sum of residues at the nontrivial zeros converges absolutely, we obtain (1).

Appendix B. Certified constants

All constants were computed with the ball arithmetic of Arb [7] through python-flint at 128 bits; every reported digit is guaranteed. The prime sum $G(10^6)$ is a sum of 78 498 terms evaluated in balls. The values $\zeta'/\zeta(1+u)$ and $\psi(1+u/2)$ come from the Arb series and digamma functions. The quantities entering Theorem 5.2 are given below.

	$B = 7/8$	$B = 11/12$
$q_0,\ y_0,\ \ell,\ u$	$2^{40},\ 5 \cdot 10^6,\ 14,\ 0.1$	$2^{40},\ 1.8 \cdot 10^9,\ 20,\ 0.08$
k	8	12
$S(q_0, u, B)$	323.75077…	587.32747…
c_0	8.61975…	11.10846…
E_0	0.2378758…	0.2560657…
$3/\sqrt{y_0}$	0.0013416…	0.0000707…
κ_0	−4.1539838…	−8.5573459…
C_-	0.7255435…	0.7140854…
C_+	2.2671750…	2.3035538…
lower bound for $L(1, \chi)$ at q_0	0.0323543…	0.0228062…
lower bound for $h(d)$, $\lvert d\rvert \geq q_0$	10 798.96…	7 612.07…

Here $G(10^6) = 0.18198181652902\ldots$, $B_1 = 0.26149721284764\ldots$ and $\gamma = 0.57721566490153\ldots$.

Appendix C. Class numbers up to one thousand

The table lists, for selected h, the number $F(h)$ of imaginary quadratic fields of class number h and the largest $|d|$ among them. The values for every $h \leq 1000$ are in the ancillary file `classnumbers_1000.txt`, together with the program that produces them.

h	$F(h)$	max $\lvert d\rvert$	h	$F(h)$	max $\lvert d\rvert$	h	$F(h)$	max $\lvert d\rvert$
1	9	163	28	457	126 043	400	13 823	32 492 923
2	18	427	29	83	166 147	425	845	35 307 787
3	16	907	30	255	134 467	450	3 745	40 868 683
4	54	1 555	31	73	133 387	475	961	43 078 963
5	25	2 683	32	708	164 803	500	8 023	53 698 747
6	51	3 763	33	101	222 643	525	1 630	51 065 683
7	31	5 923	34	219	189 883	550	3 254	69 758 203
8	131	6 307	35	103	210 907	575	1 126	61 595 227
9	34	10 627	36	668	217 627	600	21 632	72 789 403
10	87	13 843	37	85	158 923	625	1 140	81 289 723
11	41	15 667	38	237	289 963	650	3 683	85 527 187
12	206	17 803	39	115	253 507	675	1 973	104 367 187
13	37	20 563	40	912	260 947	700	12 611	96 016 243
14	95	30 067	50	345	389 467	725	1 301	106 911 667
15	68	34 483	75	237	916 507	750	5 750	113 620 123
16	322	31 243	100	1 736	1 856 563	775	1 443	134 009 467
17	45	37 123	125	290	2 944 363	800	34 149	129 667 987
18	150	48 427	150	1 233	4 163 443	825	2 305	130 788 067
19	47	38 707	175	450	4 742 467	850	4 633	135 386 227
20	350	58 507	200	5 081	7 206 763	875	1 801	149 457 163
21	85	61 483	225	709	9 119 203	900	21 669	180 619 363
22	139	85 507	250	1 472	12 208 267	925	1 579	164 446 027
23	68	90 787	275	628	14 420 323	950	5 087	186 089 443
24	511	111 763	300	6 744	19 184 323	975	2 638	184 039 507
25	95	93 307	325	691	19 385 683	1000	26 378	220 287 043
26	190	103 027	350	2 258	24 669 283			
27	93	103 387	375	1 099	26 809 843			

Declarations

Data availability. The programs (interval-arithmetic bounds in Python with python-flint, and the reduced-form counter in C with OpenMP) and the output for $h \le 1000$ are included as ancillary files.

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Use of artificial intelligence. AI coding assistants were used to write and test the computer programs and to check the bibliography. All mathematical arguments were written and verified by the author, who takes full responsibility for the content.

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Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India

Formerly, University of Calcutta (Heramba Chandra College), 23/49 Gariahat Road, Golpark, Kolkata 700029, West Bengal, India

Email address: `d.bhattacharjee@erl-forschung.de`
Email address: `itsdeep@live.com`
Email address: `egspl@proton.me`
URL: `https://orcid.org/0000-0003-0466-750X`
URL: `https://github.com/creelie`